



Shore School

2016

Year 11

Preliminary Task 4

Yearly Examination

Mathematics

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black pen
- Board approved calculators may be used
- Answer Questions 1 – 10 on the Multiple Choice answer sheet provided
- Start each of Questions 11 – 14 in a new writing booklet
- In Questions 11–14, show relevant mathematical reasoning and/or calculations
- Write your examination number on the front cover of each booklet to be handed in
- If you do not attempt a question, submit a blank booklet marked with your examination number and “N/A”
- A BOSTES reference sheet is provided.

Note: Any time you have remaining should be spent revising your answers.

Examination Number:

Set:

Total marks – 70

Section I

Pages 2–5

10 marks

- Attempt Questions 1–10
- Allow about 15 minutes for this section

Section II

Pages 6–10

60 marks

- Attempt Questions 11–14
- Allow about 1 hour and 45 minutes for this section

Section I

10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

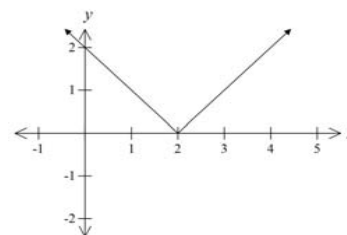
Use the multiple-choice answer sheet for Questions 1–10.

1 What is 0.00412248 written in scientific notation, correct to 4 significant figures?

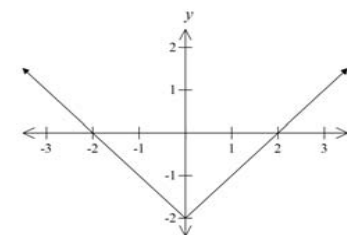
- (A) 4.1225×10^{-2}
- (B) 4.122×10^{-2}
- (C) 4.1225×10^{-3}
- (D) 4.122×10^{-3}

2 Which graph best represents $y = |x - 2|$?

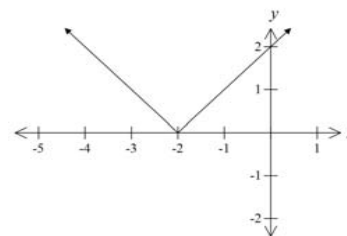
(A)



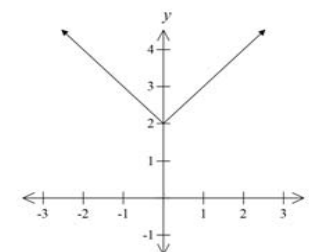
(B)



(C)



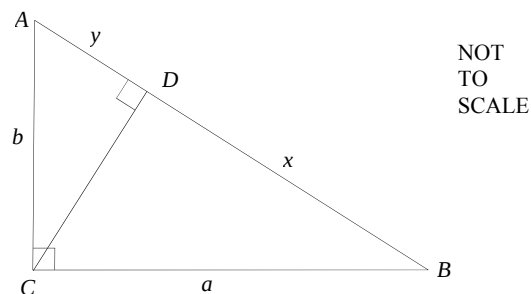
(D)



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- 3 Which of the following is equal to $\frac{1}{2\sqrt{5}-\sqrt{3}}$?
- (A) $\frac{2\sqrt{5}-\sqrt{3}}{7}$
- (B) $\frac{2\sqrt{5}+\sqrt{3}}{7}$
- (C) $\frac{2\sqrt{5}-\sqrt{3}}{17}$
- (D) $\frac{2\sqrt{5}+\sqrt{3}}{17}$
- 4 Which inequality defines the domain of the function $f(x) = \frac{1}{\sqrt{x+4}}$?
- (A) $x > -4$
- (B) $x \geq -4$
- (C) $x < -4$
- (D) $x \leq -4$
- 5 What is the gradient of the normal to the curve $y = 2x^3$ at the point where $x = 2$?
- (A) -24
- (B) $-\frac{1}{24}$
- (C) 24
- (D) $\frac{1}{24}$
- 6 Which equation represents the line perpendicular to $2x - 3y - 8 = 0$, passing through the point $(2,0)$?
- (A) $3x + 2y - 4 = 0$
- (B) $3x + 2y - 6 = 0$
- (C) $3x - 2y + 4 = 0$
- (D) $3x - 2y - 6 = 0$
- 7 What are the solutions of $2\cos x = -\sqrt{3}$ for $0^\circ \leq x \leq 360^\circ$?
- (A) $x = 30^\circ$ and 330°
- (B) $x = 60^\circ$ and 300°
- (C) $x = 150^\circ$ and 210°
- (D) $x = 120^\circ$ and 240°
- 8 What is the best description of the nature of the roots of $3x^2 - 7x + 2 = 0$?
- (A) two real, irrational roots
- (B) one real, irrational root
- (C) two real, rational roots
- (D) one real, rational root
- 9 What is the value of $\lim_{x \rightarrow -2} \frac{x^2 - 4}{x + 2}$?
- (A) Undefined
- (B) -4
- (C) 0
- (D) 4

- 10 Let $BC = a$, $AC = b$, $BD = x$ and $AD = y$. Triangle ADC is similar to triangle ACB .



Which of the following statements is correct?

- (A) $\frac{b}{y} = \frac{a}{x}$
- (B) $\frac{b}{y} = \frac{x+y}{a}$
- (C) $\frac{b}{y} = \frac{x+y}{b}$
- (D) $\frac{b}{y} = \frac{a}{b}$

Section II

60 marks

Attempt Questions 11–14

Allow about 1 hour and 45 minutes for this section

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

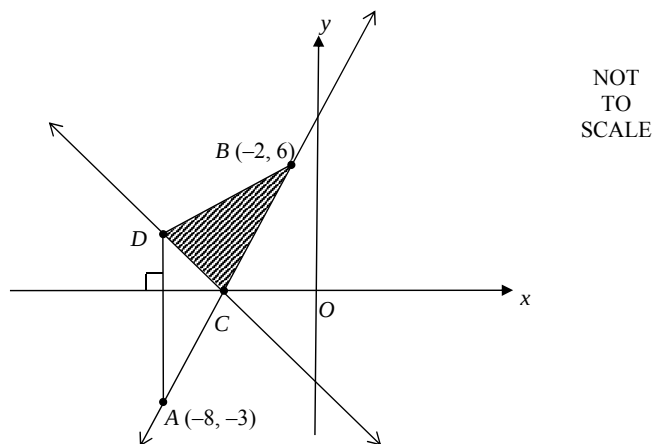
In Questions 11–14, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Use a SEPARATE writing booklet

- (a) Solve $7 - 3x \geq 22$. 2
- (b) Write, using an algebraic technique, $0.\dot{3}\dot{2}\dot{4}$ as a fraction in its simplest terms. 2
- (c) Solve $|x - 3| = 2x - 4$. 3
- (d) Simplify $\frac{x^2 - 4x - 5}{4x^2 + 4x + 4} \times \frac{x^3 - 1}{x^2 - 1}$. 3
- (e) On the number plane shade the region defined by $x^2 + y^2 < 9$ and $y \geq x + 3$. 3
- (f) A regular polygon has an interior angle of 171° .
How many sides does the polygon have? 2

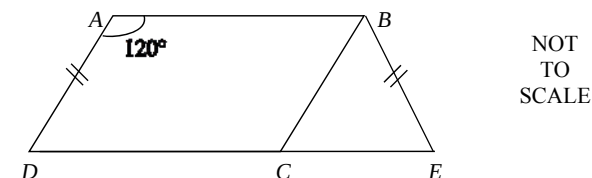
Question 12 (15 marks) Use a SEPARATE writing booklet

- (a) $A(-8, -3)$ and $B(-2, 6)$ are two points on the number plane. The line AB intersects the x -axis at C . The line CD makes an angle of 135° with the positive x -axis. The line AD is parallel to the y -axis.



- | | |
|--|---|
| (i) Given the equation of the line AB is $3x - 2y + 18 = 0$, find the co-ordinates of C . | 1 |
| (ii) Show that the equation of the line CD is $x + y + 6 = 0$. | 2 |
| (iii) Find the co-ordinates of D . | 1 |
| (iv) Find the exact length of CD , in its simplest form. | 1 |
| (v) Find the perpendicular distance from B to the line CD . | 2 |
| (vi) Find the area of $\triangle BCD$. | 1 |
| | |
| (b) Simplify $\cot \theta - \cot \theta \cos^2 \theta$. | 2 |
| (c) Show that $f(x) = x^3 - x$ is an odd function. | 2 |

- (d) $ABCD$ is a parallelogram with $\angle DAB = 120^\circ$. The side DC is produced to E so that $AD = BE$.



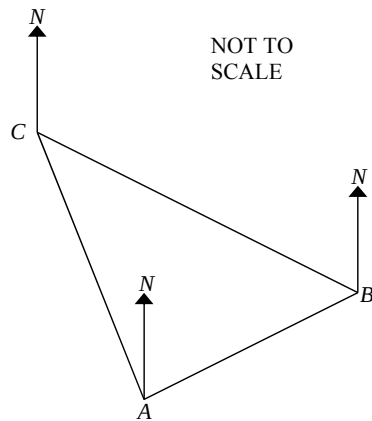
- (i) Copy or trace the diagram into your writing booklet.
- (ii) Prove that $\triangle BCE$ is an equilateral triangle.

End of Question 12

Question 12 continues on page 8

Question 13 (15 marks) Use a SEPARATE writing booklet

- (a) If α and β are roots of the equation $2x^2 + 3x - 4 = 0$, find the value of
- | | |
|--|---|
| (i) $\alpha + \beta$ | 1 |
| (ii) $\alpha\beta$ | 1 |
| (iii) $\frac{1}{\alpha} + \frac{1}{\beta}$ | 2 |
| (iv) $(\alpha - \beta)^2$ | 2 |
- (b) Find the values of A , B and C for which $2x^2 + 3x + 1 \equiv A(x+2)^2 + B(x+2) + C$. 3
- (c) A ship sails 50 km from Port A to Port B on a bearing of 063° and then sails 130 km from Port B to Port C on a bearing of 296° .



- | | |
|---|---|
| (i) Copy or trace the diagram into your writing booklet showing all the above information. | 1 |
| (ii) Show that $\angle ABC = 53^\circ$. | 1 |
| (iii) Find the distance from Port A to Port C. Give your answer correct to the nearest kilometre. | 2 |
| (iv) Find the bearing of Port A from Port C. Give your answer correct to the nearest degree. | 2 |

Question 14 (15 marks) Use a SEPARATE writing booklet

- (a) Differentiate the following with respect to x
- | | |
|------------------------------|---|
| (i) $y = (3x+4)^5$ | 2 |
| (ii) $y = \frac{x^2+5}{x-2}$ | 2 |
- (b) Differentiate $y = x^2(3-2x)^4$ with respect to x . Answer in fully factored form. 3
- (c) Find the equation of the tangent to the curve $y = x^3 + 2x^2 - 5x$ at the point $(-3, 6)$. 2
- (d) A function is defined by $f(x) = 2x^2 + 7x - 3$.
- | | |
|---|---|
| (i) Show that $f(x+h) = 2x^2 + 4xh + 2h^2 + 7x + 7h - 3$. | 1 |
| (ii) Hence, differentiate $f(x) = 2x^2 + 7x - 3$ from first principles. | 2 |
- (e) The function $f(x) = \sqrt{4x-1}$ has a tangent with gradient 2 at point N .
- | | |
|--|---|
| (i) Find the co-ordinates N . | 2 |
| (ii) Find the equation of the normal to the curve at N . | 1 |

END OF PAPER

Section 1

1. $0.00412248 = 4.122 \times 10^{-3}$ (A)

2. (A)

3.
$$\frac{1}{2\sqrt{5}-\sqrt{3}} = \frac{1}{2\sqrt{5}-\sqrt{3}} \times \frac{2\sqrt{5}+\sqrt{3}}{2\sqrt{5}+\sqrt{3}}$$

$$= \frac{2\sqrt{5}+\sqrt{3}}{4 \times 5 - 3}$$

$$= \frac{2\sqrt{5}+\sqrt{3}}{17}$$
 (D)

4. $f(x) = \frac{1}{1x+4}$

$5x+11 > 0$

$5x > -11$ (A)

5. $y = 2x^3$

$\frac{dy}{dx} = 6x^2$ at $x = 2$
 $= 6(2)^2$
 $= 24$

Gradient of the tangent = 24
 Gradient of the normal = $-\frac{1}{24}$
 as $m_1 m_2 = -1$ for perpendicular lines. (B)

6. $2x - 3y - 8 = 0$

$-3y = -2x + 8$
 $y = \frac{2}{3}x - \frac{8}{3}$
 $m_1 = \frac{2}{3}$

As $m_1 m_2 = -1$, for perpendicular lines
 $m_2 = -\frac{3}{2}$

$y - y_1 = m(x - x_1)$ (2,0) $m = \frac{3}{2}$

$y - 0 = \frac{3}{2}(x - 2)$

$2y = -3(x - 2)$

$2y = -3x + 6$

$\therefore 3x + 2y - 6 = 0$ (B)

7. $2 \cos x = -\sqrt{3}$

$\cos x = -\frac{\sqrt{3}}{2}$

Related angle $\cos 30^\circ = \frac{\sqrt{3}}{2}$

$\therefore x = 180^\circ - 30^\circ, 180^\circ + 30^\circ$ $\frac{5A}{1C}$

$= 150^\circ, 210^\circ$ (C)

8. $3x^2 - 7x + 2 = 0$

$\Delta = b^2 - 4ac$ $a=3, b=-7, c=2$

$= (-7)^2 - 4 \times 3 \times 2$

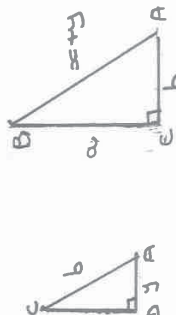
$= 25$

$\therefore$ There are 2 real, rational roots (C)

9. $\lim_{x \rightarrow -2} \frac{x^2 - 4}{x + 2} = \lim_{x \rightarrow -2} \frac{(x-2)(x+2)}{(x+2)}$

$= \lim_{x \rightarrow -2} (x-2)$

$= -4$ (B)



$\frac{b}{y} = \frac{c+y}{b}$ (C)

Question 11

a) $7 - 3x \geq 22$

$-3x \geq 15$

$x \leq -5$

b) Let $x = 0.324$

$= 0.32424 \dots$

$1000x = 324.2424 \dots$

$10x = 3.2424 \dots$

$990x = 321$

$x = \frac{321}{990}$

$x = \frac{107}{330}$

$\therefore 0.324 = \frac{107}{330}$

c) $|x - 3| = 2x - 4$

$x - 3 = 2x - 4$

$-x = -1$

$x = 1$

$3x = 7$

$x = \frac{7}{3}$

CHECK:

L.H.S. = $|x - 3|$

$= |1 - 3|$

$= 2$

R.H.S. = $2x - 4$

$= 2(1) - 4$

$= -2$

L.H.S. $\neq$ R.H.S.

$\therefore x = 1$ is not a solution

$\therefore x = \frac{7}{3}$

CHECK:

L.H.S. = $|x - 3|$

$= |2\frac{7}{3} - 3|$

$= \frac{2}{3}$

R.H.S. = $2x - 4$

$= 2(2\frac{7}{3}) - 4$

$= \frac{2}{3}$

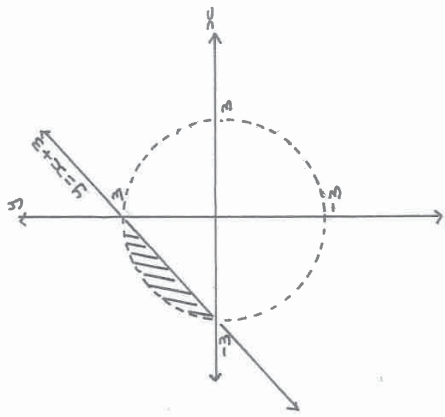
L.H.S. = R.H.S.

$\therefore x = 2\frac{7}{3}$ is a solution

d) $\frac{x^2 - 4x - 5}{4x^2 + 4x + 4} \times \frac{x - 3}{x^2 - 1}$

$= \frac{(x-5)(x+1)}{4(x^2+x+1)} \times \frac{(x-3)(x+1)}{(x-1)(x+1)}$

$= \frac{x-5}{4}$



Test (0,0) in $x^2 + y^2 < 9$ is true

$\therefore$ Shade inside the circle

Test (0,0) $y > x + 3$

$0 > 0 + 3$ is false

$\therefore$ Shade on the other side.

f) exterior = 180° - interior angle

$= 180^\circ - 171^\circ$

$= 9^\circ$

number of sides = $\frac{360}{9}$

$= 40$

OR, interior angle = $\frac{(n-2) \times 180^\circ}{n}$

$171 = \frac{(n-2) \times 180}{n}$

$171n = 180n - 360$

$-9n = -360$

$n = 40$

Question 12

a) $3x - 2y + 18 = 0$ at C, $y = 0$
 $3x = -18$
 $x = -6$

∴ The co-ordinates of C are

$(-6, 0)$

b) $m_{CD} = \tan 135^\circ$

$= -1$

$y - y_1 = m(x - x_1)$ C $(-6, 0)$

$y - 0 = -1(x + 6)$

$y = -x - 6$

∴ $x + y + 6 = 0$

iii) at D, $x = -8$

$x + y + 6 = 0$

$-8 + y + 6 = 0$

$y = 2$

∴ D is $(-8, 2)$

iv) $CD = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$
 $= \sqrt{(-6 + 8)^2 + (0 - 2)^2}$
 $= \sqrt{8}$

∴ $CD = 2\sqrt{2}$

v) $d = \left| \frac{ax_1 + by_1 + c}{\sqrt{a^2 + b^2}} \right|$ B $(-2, 6)$
 $= \left| \frac{-2 + 6 + 6}{\sqrt{1^2 + 1^2}} \right|$

$= \frac{10}{\sqrt{2}}$

$= \frac{10\sqrt{2}}{2}$

∴ $d = 5\sqrt{2}$

v) $A = \frac{1}{2}bh$
 $= \frac{1}{2} \times 2\sqrt{2} \times 5\sqrt{2}$
 $= 10$

∴ Area is 10 units^2

b) $\cot \theta - \cot \theta \cos^2 \theta = \cot \theta (1 - \cos^2 \theta)$

$= \cot \theta \times \sin^2 \theta$

$= \frac{\cos \theta}{\sin \theta} \times \sin^2 \theta$

$= \cos \theta \sin \theta$

c) $f(x) = x^3 - x$

$f(-x) = (-x)^3 - (-x)$

$= -x^3 + x$

$-f(x) = -(x^3 - x)$

$= -x^3 + x$

As $f(-x) = -f(x)$ the function is odd.



$\angle DAB = 120^\circ$ ∴ opposite angles of a parallelogram are equal

$\angle BCE = 180^\circ - 120^\circ = 60^\circ$ a straight angle is 180°

$AD = BC$ ∴ opposite sides of a parallelogram are equal

$\angle BCE = \angle BEC = 60^\circ$ In an isosceles triangle equal angles are opposite equal sides

$\angle BEC = 180^\circ - 60^\circ - 60^\circ = 60^\circ$ angle sum of a triangle is 180°

∴ $\triangle BCE$ is an equilateral triangle as all angles are 60°

Question 13

a) $2x^2 + 3x - 4 = 0$

$a = 2, b = 3, c = -4$

i) $\alpha + \beta = \frac{-b}{a}$

$= \frac{-3}{2}$

ii) $\alpha\beta = \frac{c}{a}$

$= \frac{-4}{2}$

$= -2$

iii) $\frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha + \beta}{\alpha\beta}$

$= \frac{-\frac{3}{2}}{-2}$

$= \frac{3}{4}$

iv) $(\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta$

$= \left(\frac{3}{2}\right)^2 - 4 \times -2$

$= 10\frac{1}{4}$

b) $2x^2 + 3x + 1 = A(x+2)^2 + B(x+2) + C$

$= Ax^2 + 4Ax + 4A + Bx + 2B + C$

$= Ax^2 + (4A+B)x + (4A+2B+C)$

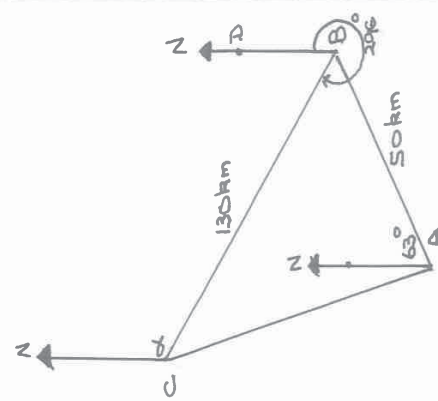
∴ $A = 2, 4A+B = 3, 4A+2B+C = 1$

$4(2)+B=3 \implies 8+B=3 \implies B=-5$

$4(2)+2(-5)+C=1 \implies 8-10+C=1 \implies C=3$

∴ $A=2, B=-5, C=3$

c) $\therefore \text{Bearing of A from C} = 116^\circ + 22^\circ = 138^\circ$



$\angle ACB = 360^\circ - 296^\circ = 64^\circ$ ∴ angles about a point add up to 360°

$\angle DBA = 180^\circ - 63^\circ = 117^\circ$ ∴ interior angles add up to 180° if the lines are parallel

$\angle ABC = 117^\circ - 64^\circ = 53^\circ$

ii) $b^2 = a^2 + c^2 - 2ac \cos B$
 $= 130^2 + 50^2 - 2 \times 130 \times 50 \cos 53^\circ$
 ≈ 11576.4047

$b \approx 107.5937019$

∴ AC is 108 km to nearest km

iii) $\cos C = \frac{a^2 + b^2 - c^2}{2ab}$

$= \frac{130^2 + 107.6^2 - 50^2}{2 \times 130 \times 107.6}$

$= 0.92857 \dots$

$C = 21.7856 \dots$

$= 22^\circ$ to the nearest deg.

$\alpha = 180^\circ - 64^\circ - 116^\circ = 1^\circ$ ∴ interior angles are equal if the lines are parallel

∴ Bearing of A from C $= 116^\circ + 22^\circ = 138^\circ$

Question 14

a) i) $y = (3x+4)^5$

$$\frac{dy}{dx} = 5(3x+4)^4 \times 3$$

$$= 15(3x+4)^4$$

ii) $y = \frac{x^2+5}{x-2}$

$$u = x^2+5 \quad v = x-2$$

$$\frac{du}{dx} = 2x \quad \frac{dv}{dx} = 1$$

$$\frac{dy}{dx} = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$$

$$= \frac{(x-2)2x - (x^2+5) \times 1}{(x-2)^2}$$

$$= \frac{2x^2 - 4x - x^2 - 5}{(x-2)^2}$$

$$= \frac{x^2 - 4x - 5}{(x-2)^2}$$

$$= \frac{(x-5)(x+1)}{(x-2)^2}$$

b) $y = x^2(3-2x)^4$

$$u = x^2 \quad v = (3-2x)^4$$

$$\frac{du}{dx} = 2x \quad \frac{dv}{dx} = 4(3-2x)^3 \times (-2)$$

$$= -8(3-2x)^3$$

$$\frac{dy}{dx} = v \frac{du}{dx} + u \frac{dv}{dx}$$

$$= (3-2x)^4 \times 2x + x^2 \times (-8(3-2x)^3)$$

$$= 2x(3-2x)^4 - 8x^2(3-2x)^3$$

$$= 2x(3-2x)^3[3-2x - 4x]$$

$$= 2x(3-2x)^3(3-6x)$$

$$= 6x(3-2x)^3(1-2x)$$

c) $y = x^3 + 2x^2 - 5x$

$$\frac{dy}{dx} = 3x^2 + 4x - 5 \quad \text{at } x = -3$$

$$= 3(-3)^2 + 4(-3) - 5$$

$$= 27 - 12 - 5$$

$$= 10$$

$$y - y_1 = m(x - x_1)$$

$$y - 6 = 10(x - (-3))$$

$$y - 6 = 10x + 30$$

$$y = 10x + 36$$

d) $f(x) = 2x^2 + 7x - 3$

$$1) f(x+h) = 2(x+h)^2 + 7(x+h) - 3$$

$$= 2(x^2 + 2xh + h^2) + 7x + 7h - 3$$

$$= 2x^2 + 4xh + 2h^2 + 7x + 7h - 3$$

ii) $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$

$$= \lim_{h \rightarrow 0} \frac{2x^2 + 4xh + 2h^2 + 7x + 7h - 3 - 2x^2 - 7x + 3}{h}$$

$$= \lim_{h \rightarrow 0} \frac{4xh + 2h^2 + 7h}{h}$$

$$= \lim_{h \rightarrow 0} \frac{h(4x + 2h + 7)}{h}$$

$$= \lim_{h \rightarrow 0} 4x + 2h + 7$$

$$\therefore f'(x) = 4x + 7$$

c) $f(x) = \sqrt{4x-1}$

$$= (4x-1)^{\frac{1}{2}}$$

$$f'(x) = \frac{1}{2}(4x-1)^{-\frac{1}{2}} \times 4$$

$$= 2(4x-1)^{-\frac{1}{2}}$$

$$= \frac{2}{\sqrt{4x-1}}$$

But $f'(x) = 2$

$$\frac{2}{\sqrt{4x-1}} = 2$$

$$\therefore \sqrt{4x-1} = 1$$

$$4x-1 = 1$$

$$4x = 2$$

$$x = \frac{1}{2}$$

$$f(x) = \sqrt{4x-1}$$

$$f\left(\frac{1}{2}\right) = \sqrt{4\left(\frac{1}{2}\right)-1}$$

$$= 1$$

$\therefore$ The co-ordinates of N are $\left(\frac{1}{2}, 1\right)$

ii) gradient of the tangent = 2
gradient of the normal = $-\frac{1}{2}$
as $m_1 m_2 = -1$ for perpendicular lines.

$$y - y_1 = m(x - x_1) \quad \left(\frac{1}{2}, 1\right)$$

$$y - 1 = -\frac{1}{2}\left(x - \frac{1}{2}\right) \quad m = -\frac{1}{2}$$

$$2y - 2 = -x + \frac{1}{2}$$

$$2x + 4y - 5 = 0 \text{ is the}$$

equation of the normal